

Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence
Topic: PEAS Framework and Task Environment Analysis
Selected System: C — Agricultural Monitoring Drone for Crop Disease Detection

Task 1: PEAS Specification

Performance Measure

Metric	Description
Disease Detection Accuracy	Percentage of correctly identified diseased vs. healthy crop patches (target: >95% precision and recall).
Coverage Efficiency	Percentage of total farmland scanned per flight mission relative to battery usage (hectares/Wh).
False Positive Rate	Frequency of healthy crops incorrectly flagged as diseased, minimized to avoid unnecessary pesticide application.
Alert Response Time	Time elapsed from disease detection to farmer notification (target: <5 minutes post-flight).
Mission Completion Rate	Percentage of planned waypoints successfully visited per flight without interruption or crash.

Environment

The agent operates over open agricultural farmland — including wheat, rice, and maize fields — typically spanning 10–500 hectares. Key external factors include variable weather conditions (wind speed, humidity, sunlight intensity), terrain irregularities, crop density, seasonal growth stages, and the presence of livestock or farm workers below the drone.

Actuators

- Quadcopter Propulsion Motors: Four variable-speed brushless motors for navigation, altitude control, and hovering.
- Precision Sprayer Nozzle (optional payload): Micro-dosing pump for targeted pesticide/fungicide application on detected disease zones.
- GPS-Guided Waypoint Navigation System: Autopilot module executing pre-programmed flight paths with real-time deviation correction.
- Wireless Telemetry Transmitter: 4G/LTE or LoRa radio for transmitting alerts, images, and GPS coordinates to the farmer's dashboard.

Sensors

- High-Resolution RGB Camera (20MP): Captures visible-spectrum imagery for canopy texture and color anomaly detection.
- Multispectral Imaging Sensor (5-band: Red, Green, Blue, Red Edge, NIR): Computes NDVI (Normalized Difference Vegetation Index) to detect chlorophyll stress invisible to the human eye.
- Thermal Infrared Camera (FLIR, 640x512 resolution): Detects heat signatures indicating fungal infection or water stress.
- GPS/GNSS Module (RTK, ±2cm accuracy): Provides precise geo-tagged location for every detected disease patch.
- Ultrasonic and LiDAR Proximity Sensors: Measures crop canopy height and maintains safe flight altitude (2–5m above crop level).
- IMU (Inertial Measurement Unit): Monitors drone orientation, wind drift, and vibration to stabilize image capture.
- Battery Voltage Sensor: Monitors remaining power to trigger return-to-home protocol before critical depletion.

Task 2: Environment Classification

Dimension	Classification	Justification
Observable	Partially Observable	The drone can only sense the area directly beneath its camera footprint at any moment; distant field sections are hidden from view.
Agent Type	Single-Agent	One drone operates independently per mission; even in a multi-drone fleet, each agent plans and acts autonomously.
Determinism	Stochastic	Wind gusts, sudden cloud cover changing lighting conditions, and random pest distribution mean the environment state changes unpredictably.

Temporality	Sequential	Detecting disease in one field patch directly influences where the drone flies next; skipping or revisiting patches
Dynamism	Dynamic	Crops grow and wind moves them continuously during the flight; lighting changes as clouds pass; the environment is constantly changing
State Space	Continuous	GPS coordinates, altitude, wind speed, camera angles, and NDVI values are all real-valued continuous variables

Task 3: Critical Analysis

Q1. Greatest Technical Challenge

The most significant technical challenge is the **Stochastic** nature of the environment. Unlike controlled lab conditions, an agricultural field is subject to constant unpredictable variation — sudden wind causing camera blur, shifting cloud shadows altering spectral reflectance values, or moisture on crop leaves changing thermal readings. These factors directly degrade the reliability of the drone's disease detection model, because a deep learning classifier trained on clear-sky, calm-wind imagery may produce incorrect predictions under different atmospheric conditions. Engineers must invest heavily in data augmentation, domain adaptation techniques, and ensemble models to build robustness — a much harder problem than handling a deterministic environment where the same input always yields the same output.

Q2. Utility Function and Trade-off Analysis

A key trade-off in this system is **Coverage Speed vs. Detection Accuracy**: flying faster covers more farmland per charge but reduces image sharpness and increases missed detections. The proposed utility function is:

$$U = \alpha * \text{DetectionAccuracy} + \beta * \text{CoverageRate} - \gamma * \text{FalsePositiveRate}$$

Where $\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$ are weighting coefficients. With this configuration, the agent balances thoroughness and efficiency — it will slow down over zones showing early disease indicators to improve accuracy, while maintaining reasonable flight speed over clearly healthy zones. **If alpha (detection accuracy weight) were doubled to 1.0**, the agent's behavior would shift dramatically: it would prioritize hovering and re-scanning suspicious patches at very low speed, accepting poor coverage rate as a trade-off. This would be ideal for high-value crops (e.g., vineyards or saffron fields) where missing a disease outbreak is catastrophically expensive, but impractical for large commodity farms where full-field coverage within one battery cycle is essential.